

September 28, 2026

Editor-in-Chief
Information Processing & Management

Dear Editor,

Please consider the enclosed manuscript, “Walk-unbiased doubly robust graph propagation: provably fair, transparent, accountable and private recommendation from exposure-biased logs”, for publication in *Information Processing & Management*.

Graph-based recommenders propagate preferences over interaction graphs that the platform’s own exposure policy has already filtered. The manuscript shows that the inverse-propensity and doubly robust corrections now used inside graph propagation are themselves biased once propagation spans more than one hop, and it gives the exact bias and a lower bound. It then proposes DRUP, an operator that is exactly unbiased and edge-wise doubly robust for any number of hops at no extra cost and without training. The same structure yields provable properties: causal exposure invariance, exact and optimal explanations, certified robustness to injected profiles, joint differential privacy and exposure-capped allocation with a certified gap. Every claim is checked on Coat, Yahoo! R3 and KuaiRec, whose test data are random or fully observed, against tuned trained baselines. The manuscript also reports where the guarantees stop holding.

We believe the work fits the journal’s interest in trustworthy information access and recommender systems. The manuscript is original, has not been published elsewhere and is not under consideration by another journal. Code and results are publicly available. In line with the double-anonymised review policy, the manuscript file contains no identifying information; author details and declarations are on the separate title page.

Sincerely,

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